

A panel analysis of Downeast English

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Agenda

Gender in language change

Introducing Downeast

Speaker sample

Acoustic measurements

Change in apparent time

Conclusion

Gender in language change

- Women lead other gender groups in language change
 - Tendency towards open and diffuse networks
 - Well-established in urban areas; not universal
- Single-variable analysis can obscure complex interactions between gender and diachronic change
 - Within variety, variables differ in their indexical values
 - Within community, gender roles differ across generations

Gender in language change

- Multivariate analyses of gender and diachronic language change focus on urban areas
 - Complications of classic gender trajectory motivated by social complexity of urban communities
- How do rural and/or rural-oriented communities differ?
 - Retraction in regional sound change
 - Compression in social complexity

Fieldsite: Downeast Maine

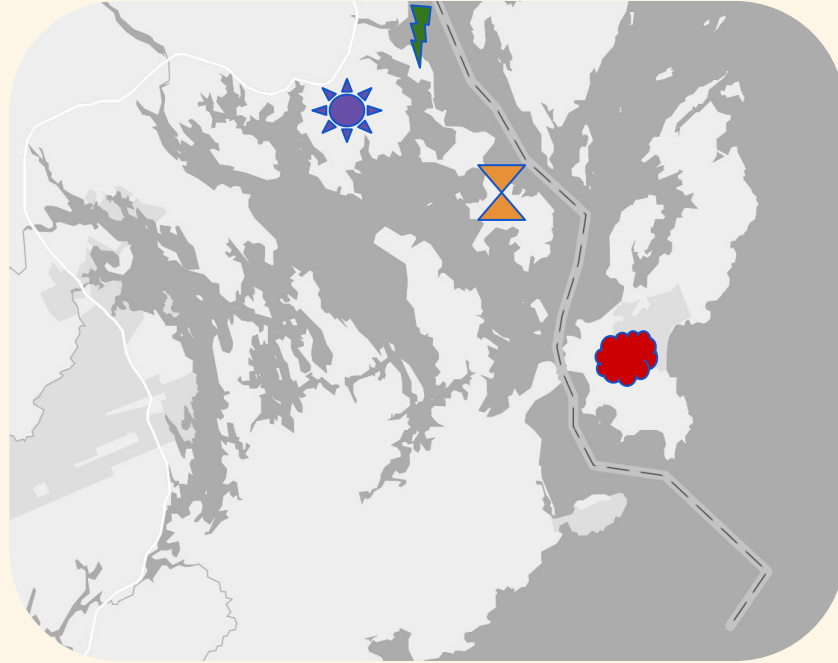
- Maine = northeasternmost state of the US
 - 2nd-most rural state in the country
- “Downeast” = northeasternmost region of Maine
 - Washington County: 3rd-least populous in the state
 - Lots of ideological negotiation over its borders!
 - Small-scale fishing, blueberry farming, tourism economy
- Eastport = easternmost city in Maine
 - Least populated city in the state

Fieldsite: Downeast Maine



Fieldsite: Eastport

- Eastport
- Passamaquoddy reservation
- “Upcountry”
- Campobello Island



Fieldsite: Eastport

- Population 1,288 in 2020
 - Peak: 5,311 in 1900
- Ancestral land of the Passamaquoddy Tribe
- Second / third largest US trading port in 19th century
- Economic collapse in 20th



Speaker sample

- 44 sociolinguistic interviews
 - 21 women
 - 23 men
- Birth years 1925-2006
 - Median = 1961
- From Eastport + beyond
 - 41 white speakers
 - 3 Passamaquoddy (Native American) speakers



Research questions

1. What are some acoustic features of Downeast English (DEE)?
2. Do three distinct DEE features reveal the same macrosocial patterning?
3. How might we explain multivariate gender trajectories across apparent time in a rural, isolated community?

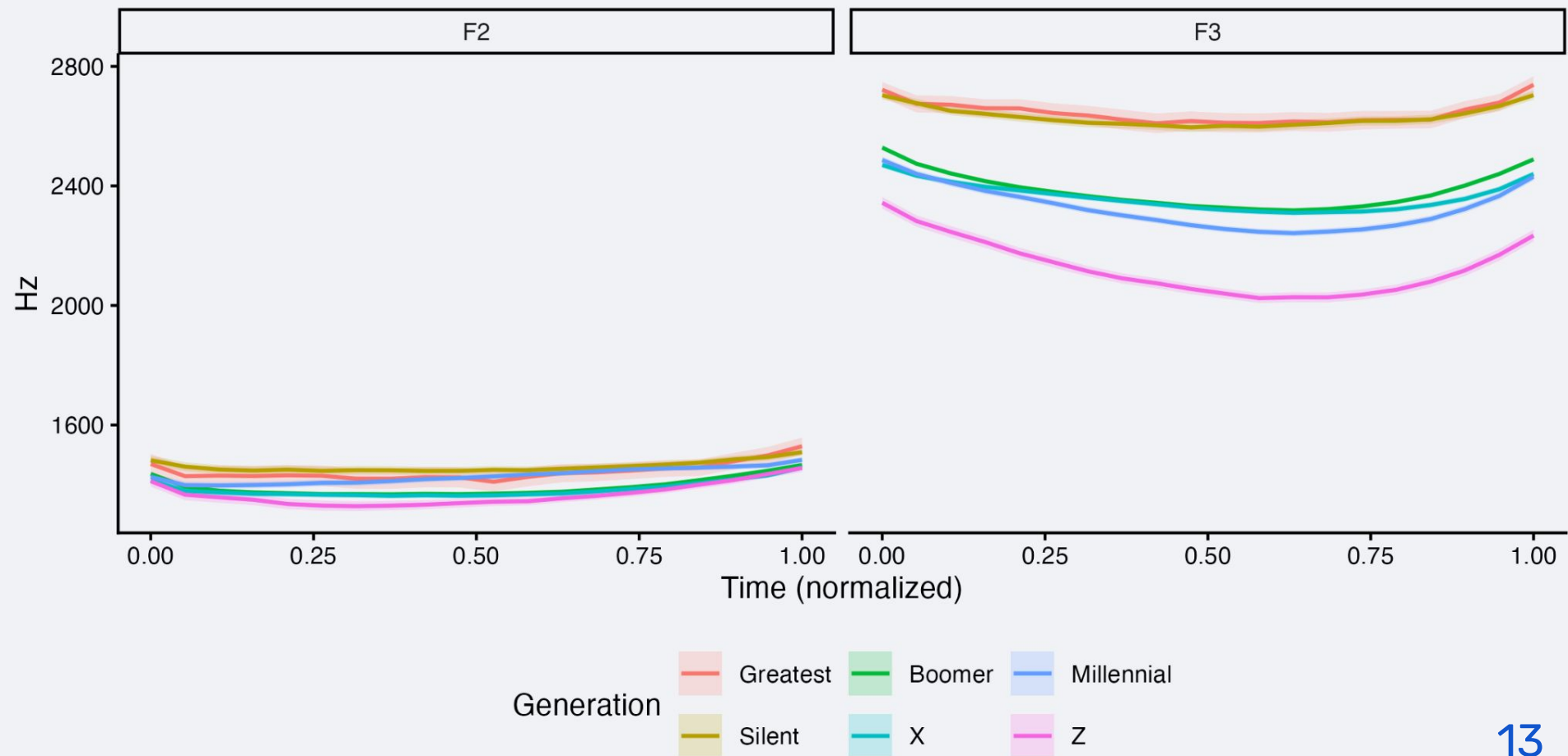
Acoustic variables

Non-prevocalic rhoticity (nPVR)

- Rhotic and vowel quality in V + R + C sequences
 - Word-internal (e.g., “Eastport”)
 - Word-final preconsonantal (e.g. “Bar Harbor”)
 - Word-final prepausal (e.g., “chowder $\emptyset$ ”)
- Shibboleth of New England speech
 - Considerable diachronic shift towards rhoticity
- Impressionistic vs. acoustic measurement
 - Acoustic: minimum F3-minus-F2 value across V + R

nPVR F2 & F3 contours by generation

Timecourse represents the vowel-rhotic sequence

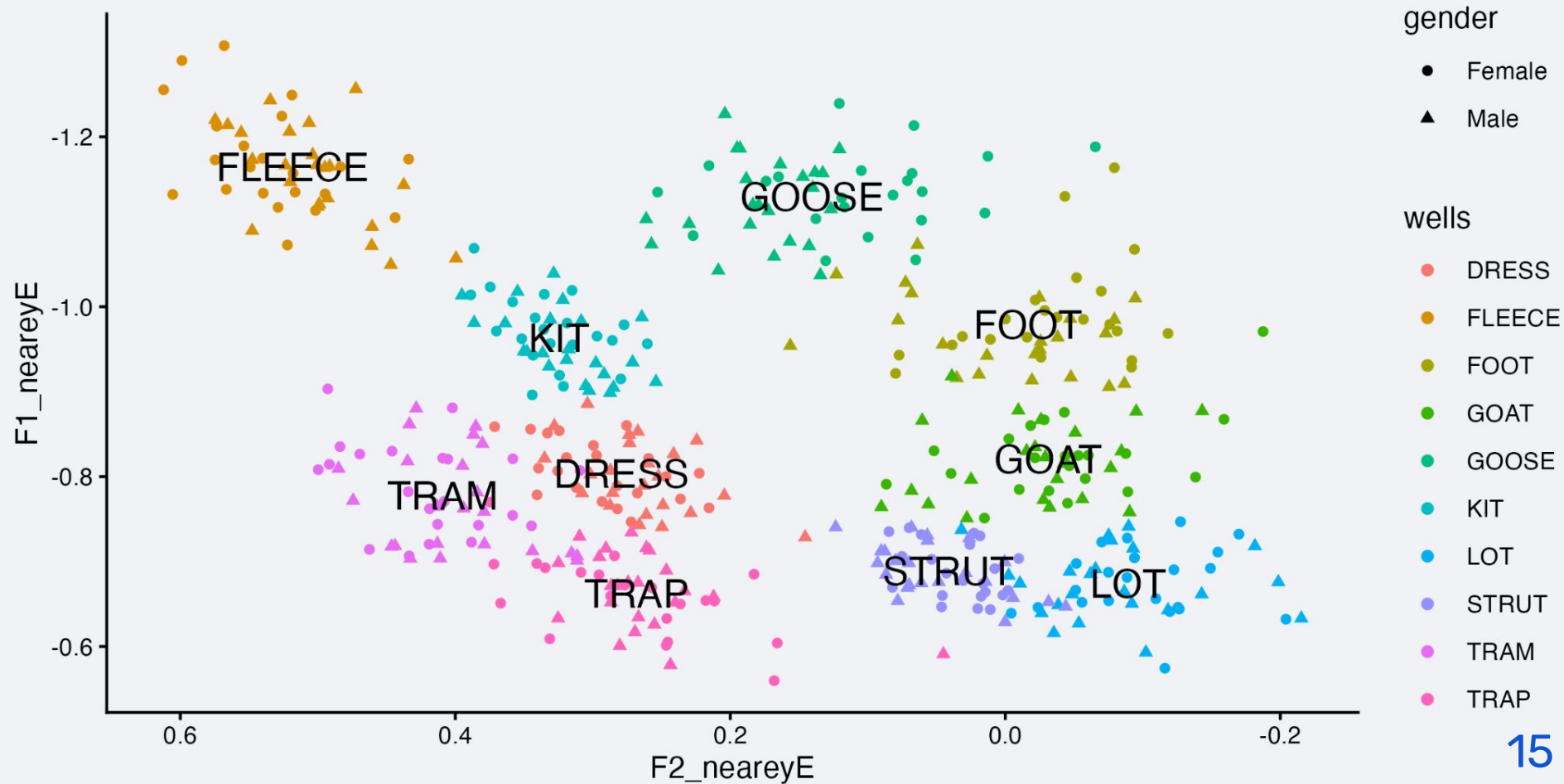


The TRAP/TRAM split

- Also referred to as a nasal short-a system
 - Tensing of /æ/ before nasals = higher and fronter
- TRAP/TRAM historically merged in New England
 - Labov and colleagues found that TRAP/TRAM uniquely distinguished primarily by F2 (frontness) in Maine
 - Fully distinct classes, but more contiguous in F1 space

Lexical class centroids

By-speaker Nearey-Extrinsic F1/F2 means



The GOOSE vowel

- Limited analysis in New England dialectology
- Work on Maritime Canadian English notes GOOSE fronting among some speakers
- Boberg analyzes GOOSE-fronting as part of the Low-Back-Merger Shift (LBMS) across Canada
 - Canadian university students

Statistical models: predictors

Shorthand	Predictor	Description
by	birth year	continuous, centered
gen	gender	factor, baseline: female, with male
lg_vow_dur	vowel duration	continuous, log transformed
lg_vr_dur	vowel + rhotic duration	continuous, log transformed
prec_poa	preceding place of articulation	factor, baseline: coronal with dorsal, glottal, labial, and other
prec_mann	preceding manner	factor, baseline: approximant with obstruent, and other
foll_poa	following place of articulation	factor, baseline: coronal with dorsal, glottal, labial, and other
foll_mann	following manner	factor, baseline: approximant with obstruent, other
foll_nas	following nasal identity	factor, baseline: other with prenasal
wells	rhotic lexical set	factor, baseline: lettER with NURSE, CURE, NEAR, NORTH/FORCE, START, and SQUARE
pos_cont	rhotic positional context	factor, baseline: word final preconsonantal with word final prepausal and word internal
spk	speaker	random intercept
wd	word type	random intercept

Statistical models: tokens and formulas

Model	Exclusions	Tokens	Formula
1. GOOSE F2	function words, discourse markers, and tokens preceding liquids	4,677	$\sim \text{lg_vow_dur} + \text{prec_poa} + \text{prec_mann} + \text{foll_poa} + \text{foll_mann} + \text{by*gen} + \text{spk} + \text{wd}$
2a. TRAP/TRAM F1	same as above	9,198	$\sim \text{lg_vow_dur} + \text{prec_poa} + \text{prec_mann} + \text{foll_nas} + \text{foll_nas:gen} + \text{foll_nas:by} + \text{foll_nase:gen:by} + \text{by:gen} + \text{spk} + \text{wd}$
2b. TRAP/TRAM F2	same as above	same as above	same as above
3. nPVR	function words, min F3 below 1400 Hz	18,589	$\sim \text{lg_vr_dur} + \text{wells} + \text{pos_cont} + \text{by*gen} + \text{spk} + \text{wd}$

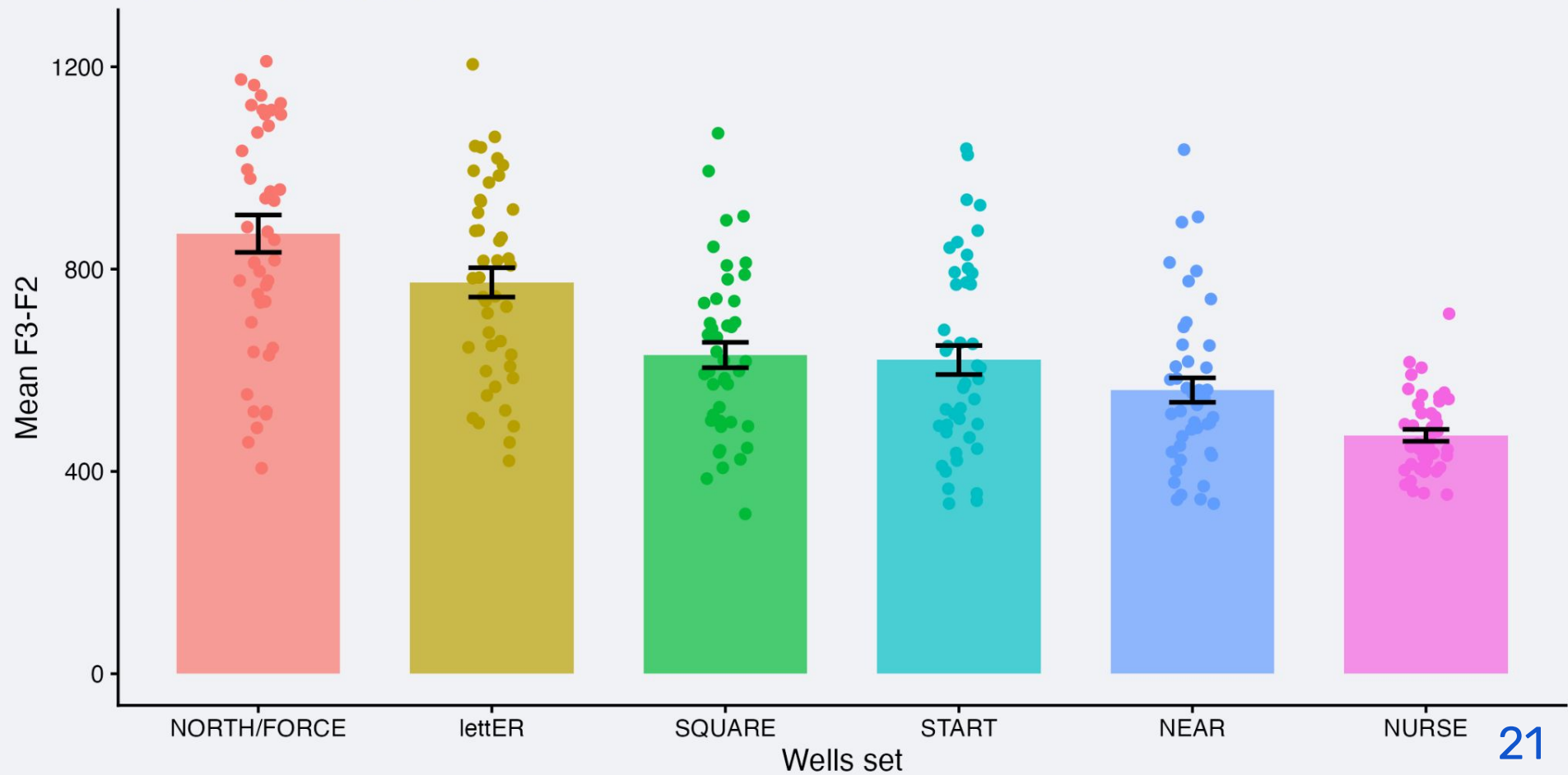
Results

Significant predictors across features

- Birth year: all
- Birth year x gender: all except TRAP/TRAM F1
- Duration: all except GOOSE F2
- Preceding POA, manner, and following POA: all
 - Not included in nPVR model

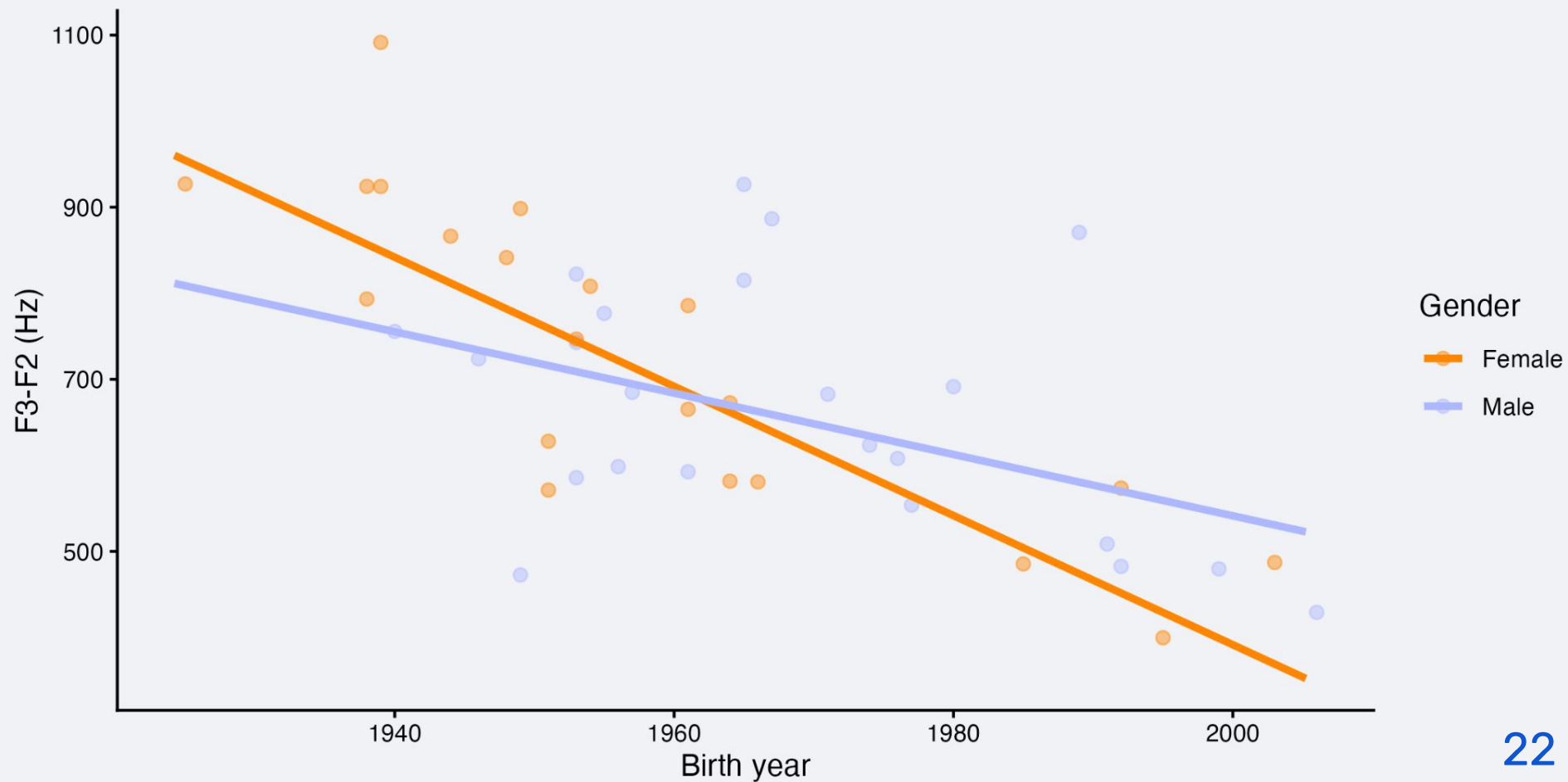
nPVR by lexical class

Points represent speaker means



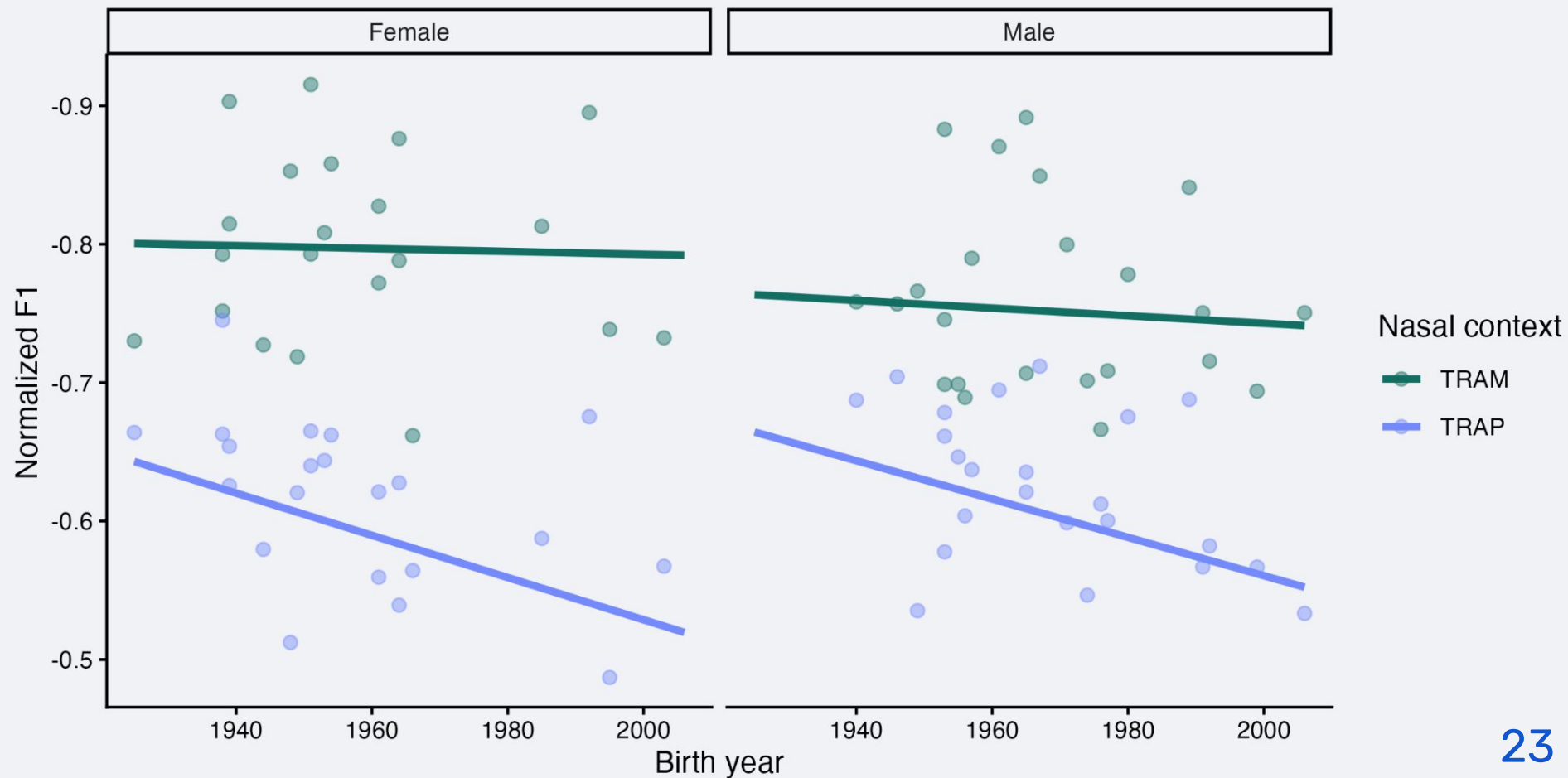
Change in rhoticity across apparent time by gender

Predicted values from linear mixed-effects model



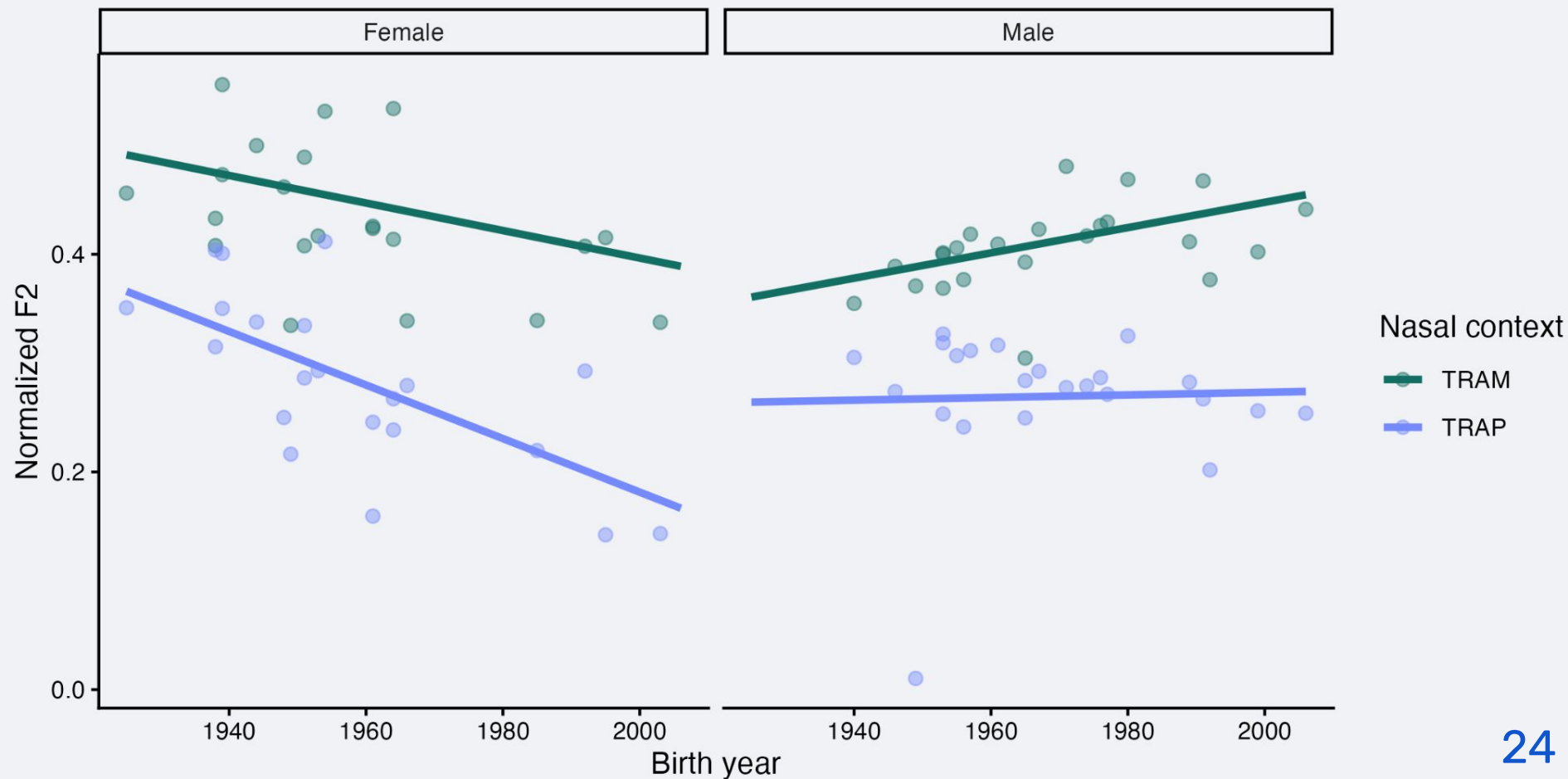
Change in TRAP/TRAM F1 across apparent time by gender

Predicted values from linear mixed-effects model



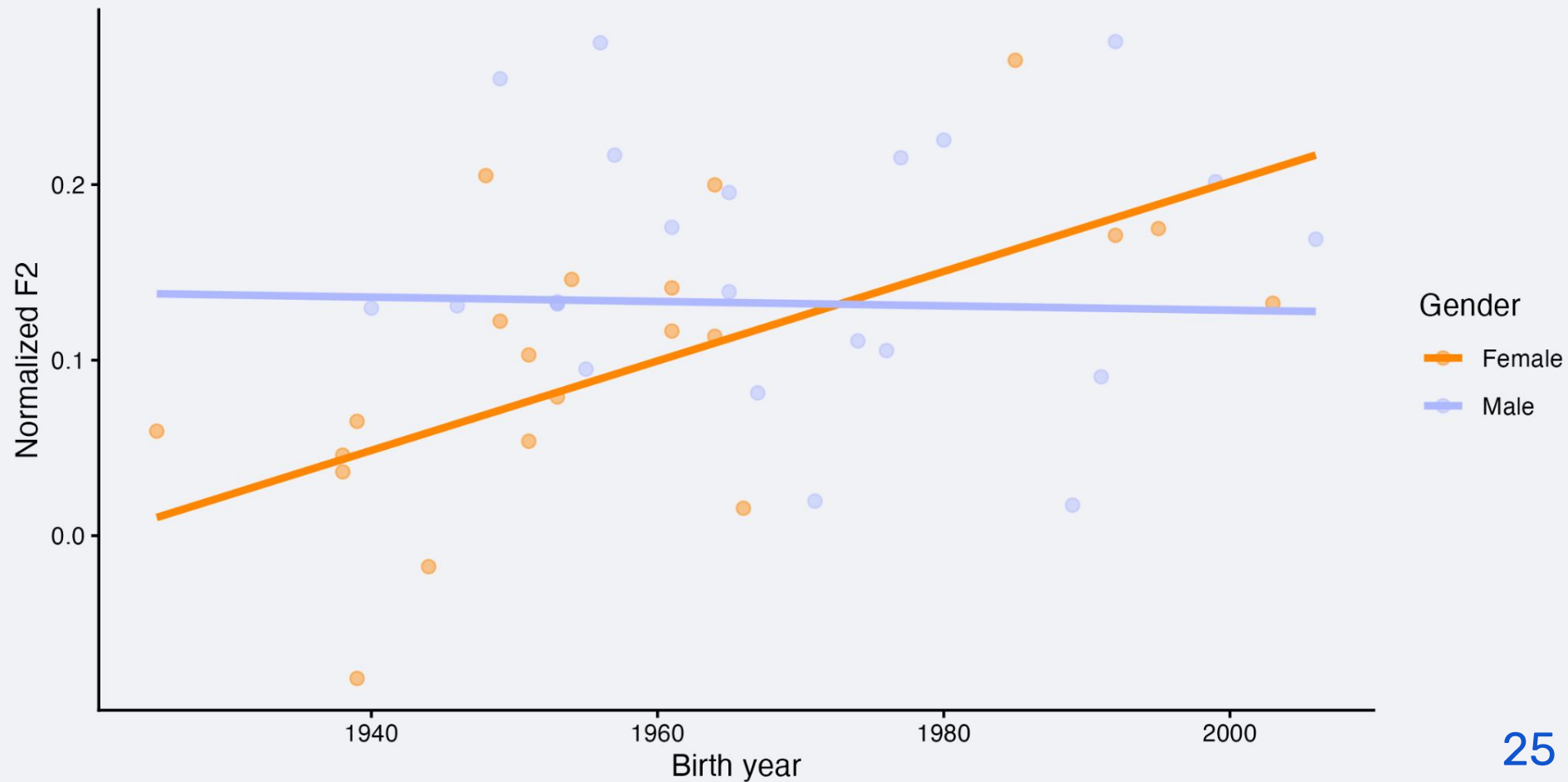
Change in TRAP/TRAM F2 across apparent time by gender

Predicted values from linear mixed-effects model



Change in GOOSE F2 across apparent time by gender

Predicted values from linear mixed-effects model



Discussion

Gender trajectories by variable

- nPVR: gross gains in rhoticity slightly led by women
- TRAP/TRAM
 - F1: TRAP lowering among both gender groups
 - Increasing distinction in F1 aligns with broader NE patterns
 - F2: women back TRAP and TRAM; men front TRAM only
 - Increasing distinction in F2 may carry some local meaning
- GOOSE: fronting among women only
 - Unclear if participation in Canadian-led shift

Characterizing trajectories of change

- Shared
 - Gains in rhoticity, led by women
 - TRAP lowering, led by women
- Divergent
 - TRAP backing only observed in women
 - GOOSE fronting only observed in women
- Inverted
 - TRAM fronting for men; backing for women

Conclusions

- nPVR, TRAM/TRAM split, and GOOSE show significant apparent time change in Downeast English
- However, these features reveal distinct gender trajectories
 - May relate to local indexical value of features
- Multivariate approach can benefit sociolinguistic analysis of all communities, not just urban areas

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Thank you!



Statistical results: nPVR

Predictor	Estimate	SE	df	t	p	Sig.
Intercept	347.648	30.999	93.3	11.21	< .001	***
Log vowel + rhotic duration	-143.994	5.284	18265.4	-27.25	< .001	***
Lexical set: NURSE	-184.998	17.231	1397.8	-10.74	< .001	***
wellsCURE	165.300	74.226	939.2	2.23	0.026	*
Lexical set: NEAR	-105.019	30.622	1355.2	-3.43	< .001	***
Lexical set: NORTH/FORCE	265.153	16.208	1732.5	16.36	< .001	***
Lexical set: START	27.533	17.851	1722.0	1.54	0.123	
Lexical set: SQUARE	-36.553	26.156	1235.7	-1.40	0.163	
Position: Word-final prepausal	-11.332	9.933	18408.0	-1.14	0.254	
Position: Word-internal	-17.129	13.179	1665.4	-1.30	0.194	
Birth year	-7.506	1.186	40.4	-6.33	< .001	***
Gender: Male	7.107	34.282	40.0	0.21	0.837	
Birth year $\times$ Male	3.942	1.759	40.1	2.24	0.031	*

Statistical results: GOOSE F2

Predictor	Estimate	SE	t	p	Sig.
Intercept	0.110	0.081	1.36	0.174	
Log duration	0.010	0.005	1.89	0.058	†
Preceding POA: Dorsal	-0.228	0.060	-3.78	< .001	***
Preceding POA: Glottal	-0.605	0.068	-8.95	< .001	***
Preceding POA: Labial	-0.457	0.023	-19.95	< .001	***
Preceding POA: Other	0.160	0.017	9.45	< .001	***
Preceding manner: Obstruent	0.247	0.017	14.41	< .001	***
Preceding manner: Other	0.339	0.019	17.54	< .001	***
Following POA: Dorsal	-0.273	0.068	-4.01	< .001	***
Following POA: Glottal	-0.482	0.188	-2.57	0.010	*
Following POA: Labial	-0.107	0.018	-5.93	< .001	***
Following POA: Other	-0.114	0.022	-5.20	< .001	***
Following manner: Obstruent	-0.065	0.077	-0.84	0.399	
Following manner: Other	-0.049	0.077	-0.64	0.519	
Birth year	0.003	0.001	3.34	0.002	**
Gender: Male	0.024	0.022	1.08	0.286	
Birth year × Male	-0.003	0.001	-2.37	0.023	*

Statistical results: TRAP/TRAM F1

Predictor	Estimate	SE	t	p	Sig.
Intercept	-0.4435	0.0146	-30.43	< .001	***
Log duration	0.0448	0.0033	13.51	< .001	***
Preceding POA: Dorsal	-0.0306	0.0118	-2.60	0.009	**
Preceding POA: Glottal	0.0386	0.0157	2.45	0.014	*
Preceding POA: Labial	0.0262	0.0088	2.97	0.003	**
Preceding POA: Other	-0.0124	0.0130	-0.95	0.343	
Preceding manner: Obstruent	-0.0830	0.0085	-9.80	< .001	***
Preceding manner: Other	-0.0718	0.0119	-6.03	< .001	***
Prenasal	-0.2126	0.0069	-30.68	< .001	***
Birth year	0.0015	0.0005	2.80	0.008	**
Gender: Male	-0.0268	0.0157	-1.70	0.097	†
Prenasal × Male	0.0704	0.0061	11.55	< .001	***
Prenasal × Birth year	-0.0014	0.0002	-6.16	< .001	***
Male × Birth year	-0.0001	0.0008	-0.17	0.863	
Prenasal × Male × Birth year	0.0003	0.0003	0.97	0.335	

Statistical results: TRAP/TRAM F2

Predictor	Estimate	SE	t	p	Sig.
Intercept	0.1843	0.0140	13.17	< .001	***
Log duration	0.0106	0.0029	3.69	< .001	***
Preceding POA: Dorsal	0.0823	0.0101	8.13	< .001	***
Preceding POA: Glottal	-0.0024	0.0135	-0.18	0.860	
Preceding POA: Labial	-0.0140	0.0076	-1.85	0.065	†
Preceding POA: Other	-0.0193	0.0112	-1.72	0.085	†
Preceding manner: Obstruent	0.1154	0.0073	15.82	< .001	***
Preceding manner: Other	0.1481	0.0103	14.44	< .001	***
Prenasal	0.1718	0.0060	28.72	< .001	***
Birth year	-0.0025	0.0006	-4.45	< .001	***
Gender: Male	-0.0017	0.0160	-0.11	0.914	
Prenasal × Male	-0.0349	0.0053	-6.58	< .001	***
Prenasal × Birth year	0.0012	0.0002	5.96	< .001	***
Male × Birth year	0.0026	0.0008	3.15	0.003	**
Prenasal × Male × Birth year	-0.0002	0.0003	-0.55	0.579	